

10 Higher-order Networks

For a paper with multiple authors, should we put a link between all possible pairs of the authors? In a social network, are the connections made at work, or school the same as family ties? Higher order networks are generalizations of the network concept beyond pairwise relations that attempt to model these situations.

Reading

Have a look at [1] for multilayer networks and [2] for hypergraphs and simplicial complexes.

10.1 Multi-layer networks

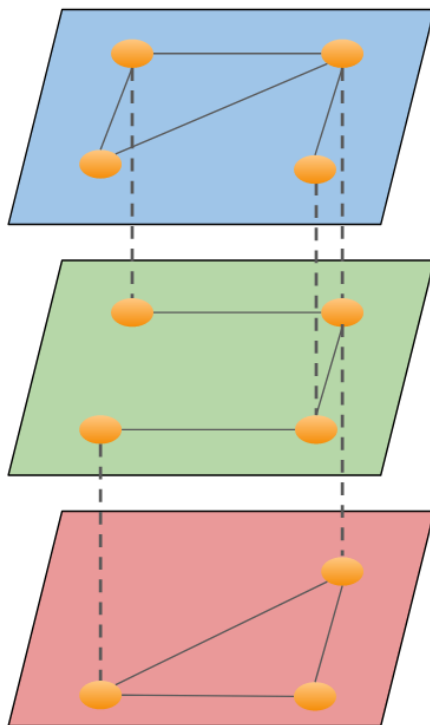


Figure 1: A multilayer network.

Perhaps the most developed generalization of a network is the idea of the **multilayer network**. The notation and terminology for multilayer networks can get quite complex and there are many different ways to express essentially the same thing. The image in Figure 1 gives the basic idea. There are nodes which are organized into **layers** where there can be connections within the layers, called **intralinks** and connections between the layers, called **interlinks**. Often the connections in the layers are of a different nature than the connections between them. Sometimes the nodes in the different layers are **replicas** i.e. the nodes in each layer represent the same entity, but may be connected differently e.g. cities connected by flights with different airlines. We will give a very brief introduction to some ideas of multi-layer networks, refer to the book of Bianconi [1] for much more detail.

10.2 Basic Definitions for Multilayer Networks

A multi-layer network is a triple $M = (Y, \vec{G}, \mathcal{G})$ where

1. Y is the set of layers, labelled $\alpha = 1, 2, 3, \dots M$
2. $\vec{G}$ is an (ordered) list of M networks, one for each layer
3. $\mathcal{G}$ is an $M \times M$ matrix of bipartite networks which give the links between layers.

So, for example, an element $G_\alpha \in \vec{G}$ is a network formed by the nodes and interlinks of layer α . An element $\mathcal{G}_{\alpha\beta}$ is a network which describes the intralinks between nodes in layer α with nodes in layer β .

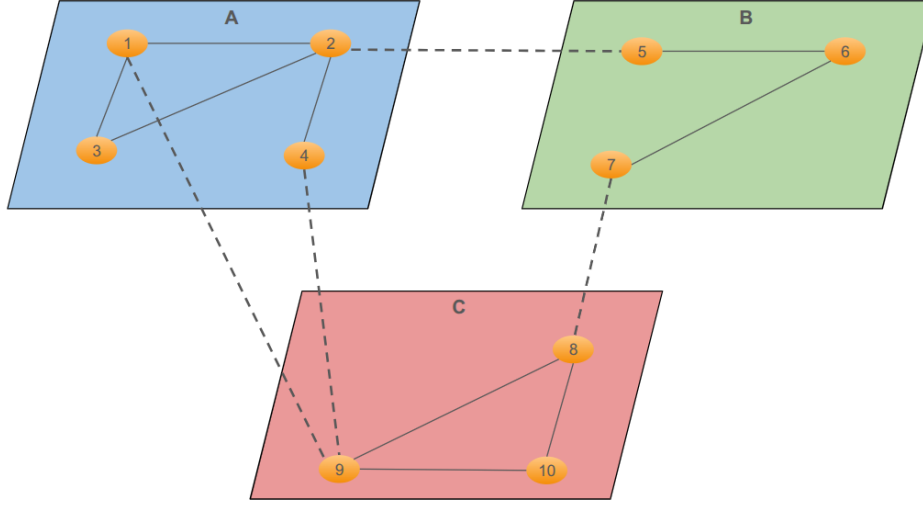


Figure 2: A multiplex network.

For the network in Figure 2 we have, representing a graph by its adjacency matrix,

$$\vec{G} = \left(\begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \right)$$

And, for example

$$\mathcal{G}_{AC} = \begin{pmatrix} & & & & 0 & 1 & 0 \\ & & & & 0 & 0 & 0 \\ & & & & 0 & 0 & 0 \\ & & & & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & & & \\ 1 & 0 & 0 & 1 & & & 0_{3 \times 3} \\ 0 & 0 & 0 & 0 & & & \end{pmatrix}$$

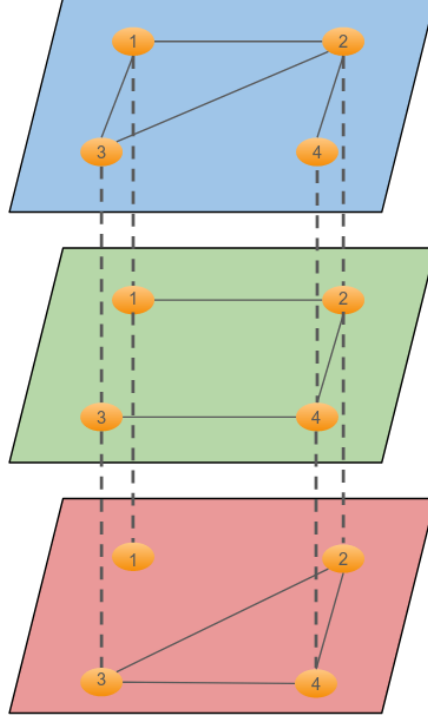


Figure 3: A multiplex network.

A **multiplex** network, shown in Figure 3 is a simplified case where

1. Nodes in each layer are identified, i.e. they are replicas.
2. Interlinks only connect the same node in each replica.

For a multiplex network we can combine all of the information into one object, first we define

$$A_{ij}^{\alpha} = \begin{cases} 1 & \text{if } i \text{ and } j \text{ are connected in layer } \alpha \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

We can then define the **supra-adjacency matrix**

$$\mathcal{A}_{i\alpha,j\beta} = \begin{cases} A_{ij}^{\alpha} & \text{if } \alpha = \beta \\ \delta_{ij} & \text{if } \alpha \neq \beta \end{cases} \quad (2)$$

By considering $i\alpha$ as a single index, taking NM values, this reduces the multiplex network to a normal one. This process is called **flattening**.

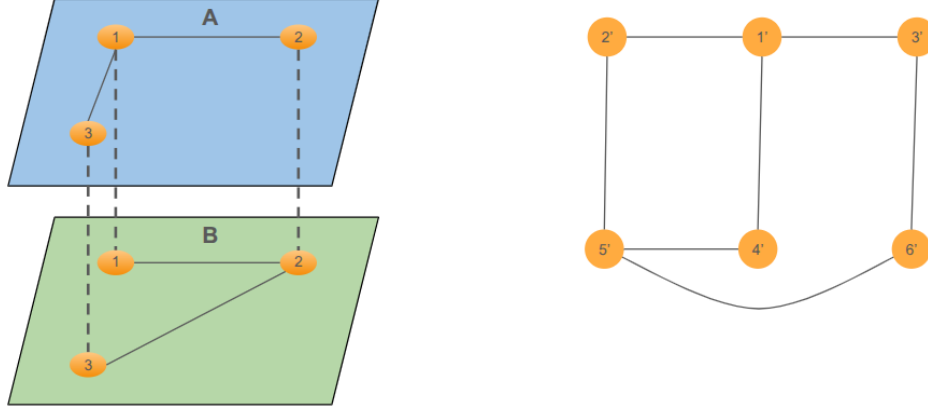


Figure 4: A simple multiplex network and its flattened version.

For the simple network on the left of Figure 4 we have

$$A^A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad A^B = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

Giving a supra adjacency matrix

$$\mathcal{A} = \begin{pmatrix} A^A & I \\ I & A^B \end{pmatrix}$$

The indices of the supra adjacency matrix are

$$\{1A, 2A, 3A, 1B, 2B, 3B\} = \{1', 2', 3', 4', 5', 6'\}$$

which gives the flattened network shown on the right of Figure 4 with adjacency matrix

$$\mathcal{A} = \begin{pmatrix} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{pmatrix}$$

Note this treats the interlinks the same as the intralinks and results in a regular network with NM nodes, however this hides the layered structure that we are interested in.

For a general multilayer network the nodes in the layers might not be replicas, indeed there can be different numbers in each layer, but we can still use the same formalism

$$A_{ij}^{\alpha\beta} = \begin{cases} 1 & \text{if } i \text{ in } \alpha \text{ linked to } j \text{ in } \beta \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

so this adjacency matrix records when nodes are connected within and between layers. Denote the number of nodes in layer α as N_α then

$$N = \sum_{\alpha} N_{\alpha}$$

and, picking some ordering for the layers, we can define a global index

$$i\alpha = i + \sum_{\alpha' < \alpha} N_{\alpha'}$$

and also define a supra-adjacency matrix $\mathcal{A}_{i\alpha,j\beta} = A_{ij}^{\alpha\beta}$, so we can flatten any multilayer network if we wish.

10.3 Degree in Multilayer Networks

In a multiplex network (same nodes in every layer) we have a **multiplex degree** for each node i

$$\vec{k}_i = (k_i^1, \dots, k_i^M) \quad (4)$$

this is a vector, and the degree distribution $P(\vec{k})$ is therefore a vector valued function. This is a bit complicated and often we want a single number associated with each node (so we can rank them, for example). There are various ways to do this. One is to consider the **aggregated network** with adjacency matrix $\tilde{A}$ where

$$\tilde{A}_{ij} = \begin{cases} 1 & \text{if } i \text{ and } j \text{ are connected in any layer} \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

Then the **aggregated degree** is

$$K_i = \sum_j \tilde{A}_{ij} \quad (6)$$

Perhaps more useful is the **link multiplicity** ν .

$$\nu_{ij} = \sum_{\alpha} A_{ij}^{\alpha} \quad (7)$$

which counts in how many layers the link (i, j) is present, this only makes sense for multiplex networks with replica nodes. Using ν we define the **overlapping degree**

$$S_i = \sum_j \nu_{ij} \quad (8)$$

Note that

$$S_i = \sum_j \nu_{ij} = \sum_j \sum_{\alpha} A_{ij}^{\alpha} = \sum_{\alpha} k_i^{\alpha}$$

is the sum of the degrees of i across each layer.

In general, for multilayer networks we can define

$$\vec{k}_{i\alpha} = (k_i^{\alpha 1}, \dots, k_i^{\alpha M}) \quad (9)$$

where

$$k_i^{\alpha\beta} = \sum_j A_{ij}^{\alpha\beta}$$

is the number of links coming out of node i that go between layers α and β . We can keep going defining degree distributions, clustering coefficients and so on, for more detail see Chapter 6 of Bianconi's book.

Note, we can think of a temporal network, especially in the sequence of networks representation, as a particular type of multilayer network, sometimes called a **multislice network**. The layers correspond to the time slices. We can either ignore the interlinks or add them as directed links between layers. To preserve the time ordering we only connect nodes in one layer with their copy in the immediately succeeding layer so we have

$$\mathcal{G}_{\alpha\beta} = \delta_{\beta, \alpha+1}$$

in a multislice representation of a temporal network. Since we have already studied temporal networks we will ignore this perspective and think mostly about the more general kinds of multi-layer networks we can build without the temporal constraints on the interlinks.

10.4 Multi-layer communities

Modularity can be generalised to the multi-layer case. In a layer the degree preserving randomization looks like

$$\frac{k_i^\alpha k_j^\alpha}{2L^\alpha}$$

where L^α is the number of edges in layer α . The labelling function also gets a layer index $c^\alpha(i)$ and multi-layer modularity is

$$\mathcal{Q} = \frac{1}{\mu} \sum_{ij\alpha\beta} \left(\left(\mathcal{A}_{i\alpha,j\beta} - \frac{k_i^\alpha k_j^\alpha}{2L^\alpha} \right) \delta_{\alpha\beta} + \omega \mathcal{A}_{i\alpha,j\beta} \delta_{ij} \right) \delta(c(i), c(j)) \quad (10)$$

Here

- μ is a normalization factor that controls the maximum value. Since we usually maximise $\mathcal{Q}$ it is not so important, we can set it so the theoretical max of $\mathcal{Q}$ is 1.
- If $\omega = 0$ this is just the average of the modularities from each layer.
- If $\omega \neq 0$ then if links are present in many layers this enhances the modularity contribution of those links.

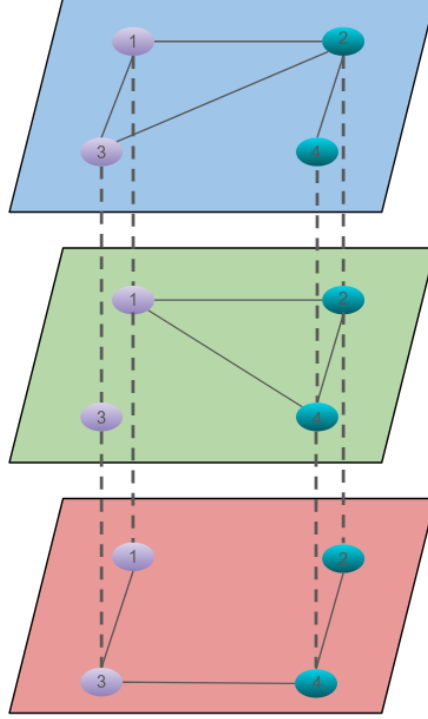


Figure 5: A multilayer network with communities suggested by colour.

Figure 5 shows a very simple example for a multiplex network, the link $\{2, 4\}$ is present in all the layers, and the link $\{1, 3\}$ in two out of three, suggesting these nodes should cluster together.

10.5 Multilayer centrality

Finally, there are many ways to generalise the centrality measures like eigenvector centrality or PageRank to multilayer networks. One interesting metric is called **versatility**. This is supposed to measure the nodes that play the strongest role in the cohesion of the multilayer structure. We denote the **eigenvector versatility** by $\theta_{i\alpha}$. This is the solution to

$$\sum_{j\beta} \mathcal{A}_{i\alpha,j\beta} \theta_{j\beta} = \lambda_1 \theta_{i\alpha} \quad (11)$$

That is, we solve the eigenvalue problem using the supra-adjacency matrix instead of the adjacency matrix. There is a similar version for Katz and

PageRank centralities. We then define

$$x_i = \sum_{\alpha} \theta_{i\alpha} \quad (12)$$

as the **node versatility** (defined only for multiplex networks where the nodes are replicas and the sum over layers makes sense!). The name versatility is justified because nodes with a high value of this index are important in multiple layers e.g. a scientist who has a large influence in their discipline might be more ‘important’ but less ‘versatile’ than a scientist who has had a moderate influence across a large number of disciplines - chemistry, physics, philosophy etc.

10.6 Hypergraphs

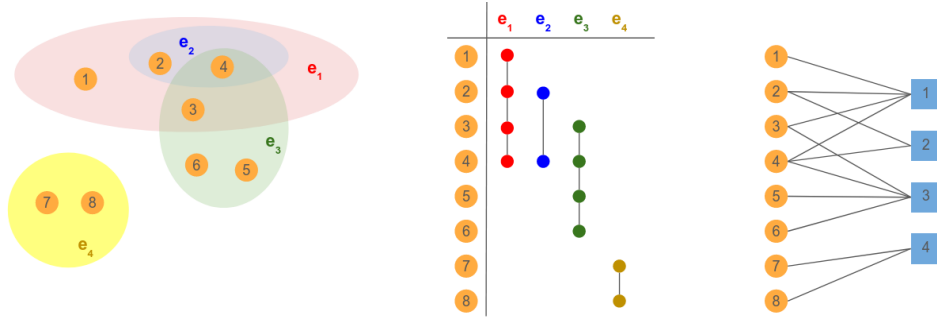


Figure 6: The same hypergraph drawn in three different ways: as overlapping sets, Parallel Aggregated Ordered Hypergraph (PAOH) and as a bipartite network.

We will give an even more rapid introduction to hypergraphs, basically just giving the definition, more details can be found in Bianconi’s book. A **hypergraph** H consists of a set of vertices V and a set of **hyperedges** E where an edge is a set of elements of V . The key difference here is that the hyperedges can link more than two entities at a time e.g. all the authors of a paper. The hypergraph in Figure 6 is

$$V = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$$E = \{\{1, 2, 3, 4\}, \{2, 4\}, \{3, 4, 5, 6\}, \{7, 8\}\}$$

There is a rich mathematical theory for hypergraphs, and many of the same sorts of theorems and algorithms which work for networks also work for

hypergraphs. We can also represent a hyper graph as a standard bipartite graph. Define a bipartite network B_H where the left part is the set of vertices of H , and the right part is the set of edges of H . Make a link from a node in V to a node in E if that node is in E . An example is shown in Figure 6. Note this generates a bipartite graph with a large number of edges. If the bipartite network is projected to a unipartite one, this gives an even larger number of edges. Hence treating the data as a hypergraph can avoid inflating the degree of nodes and instead use ‘hyper’ versions of things like degree, centrality and clustering coefficients.

10.7 Simplicial Complexes

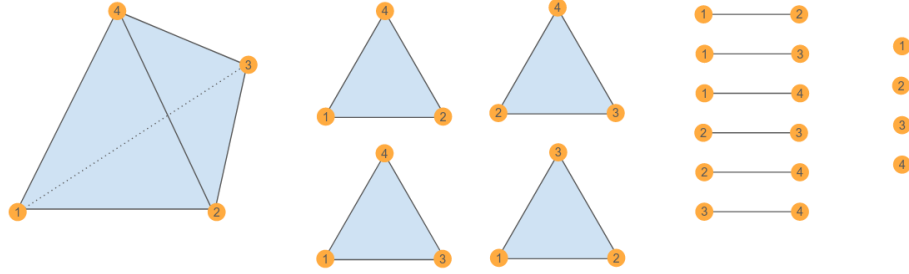


Figure 7: A tetrahedron (3-simplex) broken down into 2, 1 and 0 simplices.

A last example of a generalisation of a network is a **simplicial network** or **simplicial complex**. These are mostly studied in pure math, but are starting to have some applications in network science for analysing real data. A **simplex** is a higher dimensional generalization of a triangle, so a 2-simplex is a triangle, a 3-simplex is a tetrahedron and we keep going. A simplex is the most elementary shape in d dimensions, which joins $d + 1$ vertices and is made out of simplices of smaller dimension, see Figure 7.

More formally, a simplicial complex is a hypergraph where the hyper-edges are closed under inclusion. Spelling it out, we have a set V of nodes and simplices $\sigma^p = \{v_1, v_2, \dots, v_p\}$ which function like edges. A simplicial complex $\mathcal{K}$ is a hypergraph where

- If $\sigma \in \mathcal{K}$ and $\tau \subset \sigma$ then $\tau \in \mathcal{K}$
- $\{v\} \in \mathcal{K}$ for $v \in V$

If $\mathcal{K}$ contains the 3-simplex $\{v_0, v_1, v_2\}$ (a triangle), then it also contains $\{v_0, v_1\}, \{v_1, v_2\}, \{v_2, v_0\}$ (the sides) and $\{v_0\}, \{v_1\}, \{v_2\}$ (the points). The subsets of a simplex are called its **faces**.

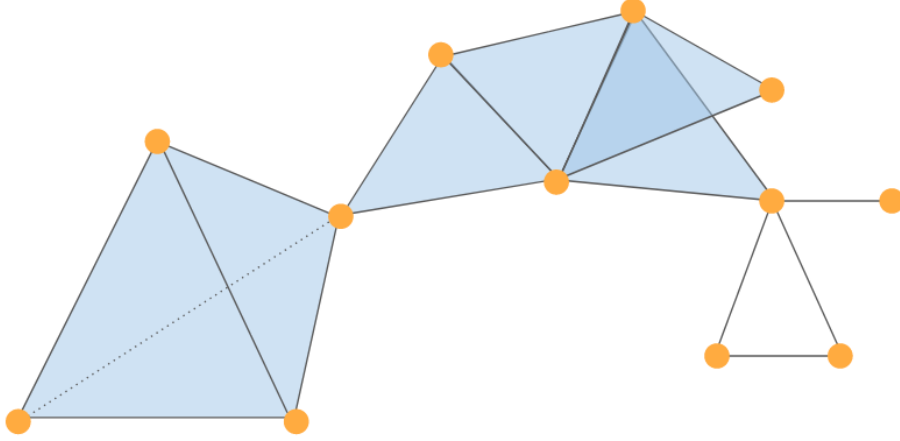


Figure 8: A simplicial complex with 3, 2 and 1 simplexes.

An example of a simplicial complex is shown in Figure 8. Simplicial complexes have a richer mathematical structure than hypergraphs and allow more interesting data analysis. Analysing simplices is called **topological data analysis** and involves coarse graining, averaging, the data at various levels and seeing what kinds of structures persist or computing topological invariants of the simplicial complex generated by the data and seeing how they change as the data changes.

The decision to use a regular network, hypergraph or simplicial complex depends on the problem. For example if we have a paper with 4 authors we might represent that in a network as an edge between every pair of authors, giving $4(3)/2 = 6$ edges and a 4-clique in the network. In a hypergraph it would be a single hyperedge $\{\text{author1}, \text{author2}, \text{author3}, \text{author4}\}$. In a simplicial complex it would be a 3-simplex (tetrahedron) together with the 2,1 and 0 simplexes that are its faces. A simplicial complex brings a lot of extra ‘baggage’ compared to a hypergraph, but this also allows the use of advanced mathematical tools.

10.8 Basics of Simplicial Complexes

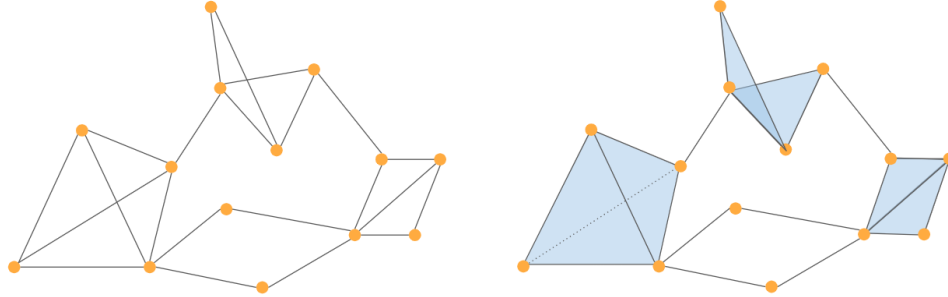


Figure 9: A network and its clique complex.

We can extract a network from a simplicial complex by simply considering only the nodes and links (0 and 1-simplices). This is called the **1-skeleton**. Conversely, we can construct a simplicial complex from any network called its **clique complex** by turning every clique of $d + 1$ nodes into a d -simplex. Figure 9 shows an example where the left is the 1-skeleton of the right and the right is the clique complex of the left.

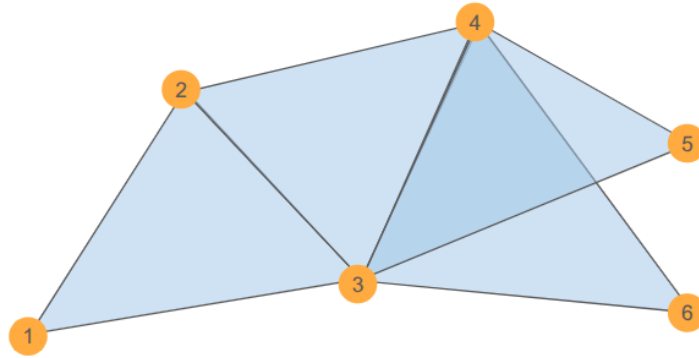


Figure 10: A simplicial complex.

The **generalized degree** $k_{d,m}(\alpha)$ is the number of d -simplices incident on the m -simplex α . This might be a bit confusing, so look at the example in Figure 10. The $k_{1,0}$ degrees are

$$k_{1,0} = (2, 3, 5, 4, 2, 2)$$

i.e., just the regular degrees. The $k_{2,0}$ degrees are

$$k_{2,0} = (1, 2, 4, 3, 1, 1)$$

i.e. the number of triangles incident at each of the nodes. $k_{2,1}$ would be the number of triangles incident at each **edge**:

e	$k_{2,1}(w)$
(1, 2)	1
(1, 3)	1
(2, 3)	2
(2, 4)	1
(3, 4)	3
(3, 5)	1
(3, 6)	1
(4, 5)	1
(4, 6)	1

The degree $k_{d,m}(\alpha)$ with $d < m$ is fully determined by the dimension of the simplex e.g. $k_{0,2}(t)$, is the number of 0-simplexes (nodes) incident on the 2-simplex (triangle) t . This is always 3!

Note, the complex in Figure 10 is made up completely of triangles. A **pure** d -dimensional simplicial complex is one made of a set of d -dimensional simplices and their faces. In this case we can define something similar to the adjacency matrix, called the **adjacency tensor**. The adjacency tensor of a simplicial complex K is

$$A_{\alpha}^d = \begin{cases} 1 & \text{if } \alpha \in S_d(K) \\ 0 & \text{otherwise} \end{cases} \quad (13)$$

Here $S_d(K)$ is the set of all d -simplices of the complex and the index α ranges over $Q_d(N)$, all the d -simplices which would *possibly* be generated from N nodes e.g. $Q_1(N)$ consists of $N(N-1)/2 = \binom{N}{2}$ possible edges. In the same way we can get the degrees in a network by summing the rows of the adjacency matrix we have

$$k_{d,m}(\alpha) = \sum_{\beta \in Q_d(N), \beta \subseteq \alpha} A_{\beta}^d \quad (14)$$

to get the degree. There are even more relations like this, as well as analogues of the Laplacian which ultimately allow you to count things like the number of ‘holes’ in a complex. The book [2] gives many more details.

References

- [1] Ginestra Bianconi. *Multilayer networks: structure and function*. Oxford university press, 2018.
- [2] Ginestra Bianconi. *Higher-order networks*. Cambridge University Press, 2021.